

Highlights

Wigner-Eckart Factorization of the Polyatomic Boltzmann Collision Operator

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- Wigner-Eckart factorization extended to the polyatomic Borgnakke-Larsen model.
- Collision integral reduced to a 9D kinematic core via $SO(3)$ rotational symmetry.
- Macroscopic angular geometry decouples from continuous internal energy exchange.
- Tailored coordinate transformations enable spectral quadrature for the 9D operator.
- Null-space embedding natively conserves mass, momentum, and total molecular energy.

Wigner-Eckart Factorization of the Polyatomic Boltzmann Collision Operator

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ABSTRACT

We extend the Wigner–Eckart factorization to the continuous Borgnakke–Larsen model for polyatomic gases. By tensorizing a polar velocity-space basis with internal energy polynomials, the macroscopic angular geometry is decoupled from the underlying scattering physics. Exploiting the SO(3) rotational invariance of binary collisions reduces the multidimensional collision integral to a nine-dimensional kinematic core. This core isolates the continuous exchange of kinetic and internal energies. A singular quadrature strategy is introduced to resolve the kinematic singularities arising from the energy partitioning, enabling spectrally convergent integration. The decomposition yields a sparse geometric tensor and a dense physical interaction tensor, substantially reducing memory and arithmetic complexity while explicitly preserving the macroscopic conservation of mass, momentum, and total energy.

1. Introduction

2. Background and Notation

In this section, we establish the mathematical foundation for the spatially homogeneous Boltzmann equation governing polyatomic gases. We define the continuous Borgnakke–Larsen collision operator, outline its invariant measure and asymmetric weak form, introduce the tensorized spectral Galerkin method, and formulate the discrete collision tensor.

Throughout this paper, boldface letters denote vectors in $\mathbb{R}^3$ (e.g., $\mathbf{v}$, $\mathbf{w}$), their non-bold counterparts denote scalar magnitudes or speeds (e.g., $v = |\mathbf{v}|$, $w = |\mathbf{w}|$), and hats denote unit directional vectors (e.g., $\hat{\mathbf{v}} = \mathbf{v}/v$). Primed variables designate post-collision states (e.g., $\mathbf{v}'$, I').

2.1. Continuous Polyatomic Collision Model

We consider the initial value problem for the spatially homogeneous Boltzmann equation,

$$\frac{\partial f(t, \mathbf{v}, I)}{\partial t} = Q(f, f)(t, \mathbf{v}, I), \quad t \in \mathbb{R}^+, \quad \mathbf{v} \in \mathbb{R}^3, \quad I \in [0, \infty), \quad (1)$$

which describes the time evolution of the distribution function f . Each particle state is characterized by its translational velocity $\mathbf{v}$ and microscopic internal energy I .

For a colliding pair with pre-collision states $(\mathbf{v}, I)$ and $(\mathbf{w}, J)$, the relative velocity is $\mathbf{u} = \mathbf{v} - \mathbf{w}$ with magnitude u . Elastic interactions conserve total momentum and total molecular energy. The total collision energy is

$$E = \frac{1}{4}u^2 + I + J = \frac{1}{4}(u')^2 + I' + J'. \quad (2)$$

Following the Borgnakke–Larsen parameterization, the post-collision states are governed by the scattering direction $\hat{\mathbf{u}}' \in \mathbb{S}^2$ and two scalar partition parameters, $R, r \in [0, 1]$. The parameter R dictates the fraction of the total energy converted into kinetic energy, yielding the post-collision relative velocity $\mathbf{u}' = 2\sqrt{RE}\hat{\mathbf{u}}'$, which establishes

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the polar deflection angle $\cos \chi = \hat{\mathbf{u}} \cdot \hat{\mathbf{u}}'$. The parameter r distributes the remaining internal energy between the two interacting particles. The post-collision states are uniquely determined by

$$\mathbf{v}' = \frac{1}{2}(\mathbf{v} + \mathbf{w} + \mathbf{u}'), \quad \mathbf{w}' = \frac{1}{2}(\mathbf{v} + \mathbf{w} - \mathbf{u}'), \quad (3a)$$

$$I' = r(1 - R)E, \quad J' = (1 - r)(1 - R)E. \quad (3b)$$

The polyatomic collision operator corresponds to the multi-dimensional integral

$$Q(f, f)(\mathbf{v}, I) = \int_{\mathbb{R}^3} \int_0^\infty \int_0^1 \int_0^1 \int_{\mathbb{S}^2} B \left[f(\mathbf{v}', I') f(\mathbf{w}', J') \left(\frac{IJ}{I'J'} \right)^\nu - f(\mathbf{v}, I) f(\mathbf{w}, J) \right] d\Phi, \quad (4)$$

where $B \equiv B(u, \cos \chi, I, J, R, r)$ is the physical transition probability rate. The differential measure $d\Phi = \varphi_\nu(r) \psi_\nu(R) (1 - R) R^{1/2} d\hat{\mathbf{u}}' dR dr dJ d\mathbf{w}$ incorporates the Jacobian of the collision transformation. The partition functions $\varphi_\nu(r) = (r(1 - r))^\nu$ and $\psi_\nu(R) = (1 - R)^{2\nu}$ dictate the energy exchange. The constant $\nu = (D - 5)/2$ relates the microscopic collision mechanics to the macroscopic internal degrees of freedom D of the gas.

2.2. Micro-Reversibility and the Asymmetric Weak Form

To ensure micro-reversibility, the collision integral relies on a fully invariant measure dA (?). This measure natively absorbs the pre-collision density of states $(IJ)^\nu$ to neutralize the asymmetry of the collision transformation:

$$dA = B \varphi_\nu(r) \psi_\nu(R) (1 - R) R^{1/2} (IJ)^\nu dR dr d\hat{\mathbf{u}}' dJ d\mathbf{w} dI d\mathbf{v}. \quad (5)$$

Multiplying the strong form by a test function $\zeta(\mathbf{v}, I)$ and integrating over the target state space yields the asymmetric weak form of the collision operator:

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^+} Q(f, f) \zeta dI d\mathbf{v} = \int_{(\mathbb{R}^3)^2 \times (\mathbb{R}^+)^2 \times [0,1]^2 \times \mathbb{S}^2} \frac{f(\mathbf{v}, I) f(\mathbf{w}, J)}{(IJ)^\nu} [\zeta(\mathbf{v}', I') - \zeta(\mathbf{v}, I)] dA. \quad (6)$$

Substituting the explicit definition of dA into this weak form analytically cancels the asymmetric quotient $(IJ)^\nu$ in the denominator. This reduces the integral to the uncorrupted transition measure:

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^+} Q(f, f) \zeta dI d\mathbf{v} = \int \left[\zeta(\mathbf{v}', I') - \zeta(\mathbf{v}, I) \right] f(\mathbf{v}, I) f(\mathbf{w}, J) B \varphi_\nu(r) \psi_\nu(R) (1 - R) R^{1/2} dR \dots d\mathbf{v}. \quad (7)$$

2.3. Tensorized Spectral Galerkin Bases

Spectral methods approximate the continuous distribution function by projecting it onto a truncated basis of orthogonal polynomials. We tensorize a three-dimensional velocity-space basis (associated Laguerre polynomials and spherical harmonics) with orthogonal polynomials for the internal energy space.

Defining a composite spectral state index $\alpha = (k, l, m, i)$, the orthogonal basis functions decouple the radial, angular, and internal energy dependencies:

$$\phi_\alpha(\mathbf{v}, I) = R_{k,l}(v) Y_{l,m}(\hat{\mathbf{v}}) H_i(I). \quad (8)$$

The distribution function is expanded using time-dependent spectral coefficients $c_\alpha(t)$ weighted by the absolute equilibrium states:

$$f(t, \mathbf{v}, I) \approx \tilde{M}(v) \tilde{E}(I) \sum_{\alpha=1}^N c_\alpha(t) \phi_\alpha(\mathbf{v}, I), \quad (9)$$

where N is the total number of basis functions. The reference Maxwellian weight $\tilde{M}(v)$ is defined by the reference density $\tilde{\rho}$ and temperature $\tilde{T}$:

$$\tilde{M}(v) = \frac{\tilde{\rho}}{(2\pi\tilde{T})^{3/2}} \exp\left(-\frac{v^2}{2\tilde{T}}\right). \quad (10)$$

Crucially, the theoretical equilibrium distribution for the continuous polyatomic model, $\tilde{E}(I)$, inherently contains the I^ν density of states:

$$\tilde{E}(I) = \frac{1}{\tilde{T}^{\nu+1} \Gamma(\nu+1)} I^\nu \exp\left(-\frac{I}{\tilde{T}}\right). \quad (11)$$

2.4. Macroscopic Collision Tensor

Applying the Galerkin method transforms the continuous integro-differential equation into a coupled system of ordinary differential equations. Setting the test function $\zeta = \phi_{\alpha_1}(\mathbf{v}, I)$ and substituting the spectral ansatz into the reduced weak form (7) yields:

$$\frac{\partial c_{\alpha_1}}{\partial t} = \sum_{\alpha_2=1}^N \sum_{\alpha_3=1}^N C_{\alpha_1, \alpha_2, \alpha_3} c_{\alpha_2} c_{\alpha_3}. \quad (12)$$

The temporal evolution is governed by the macroscopic Cartesian collision tensor C , which strictly separates into a positive gain and negative loss component $C = C^+ - C^-$. Because the reference weights $\tilde{E}(I)\tilde{E}(J)$ naturally provide the $I^\nu J^\nu$ terms required for detailed balance, the discrete tensors assemble without the need for artificial micro-reversibility quotients.

The exact Cartesian tensors evaluate to twelve-dimensional integrals over the absolute laboratory frame:

$$C_{\alpha_1, \alpha_2, \alpha_3}^+ = \int_{(\mathbb{R}^3)^2 \times (\mathbb{R}^+)^2 \times [0,1]^2 \times \mathbb{S}^2} B \phi_{\alpha_1}(\mathbf{v}', I') \phi_{\alpha_2}(\mathbf{v}, I) \phi_{\alpha_3}(\mathbf{w}, J) d\tilde{\Phi}, \quad (13a)$$

$$C_{\alpha_1, \alpha_2, \alpha_3}^- = \int_{(\mathbb{R}^3)^2 \times (\mathbb{R}^+)^2 \times [0,1]^2 \times \mathbb{S}^2} B \phi_{\alpha_1}(\mathbf{v}, I) \phi_{\alpha_2}(\mathbf{v}, I) \phi_{\alpha_3}(\mathbf{w}, J) d\tilde{\Phi}, \quad (13b)$$

where the combined integration measure $d\tilde{\Phi}$ absorbs the exact reference weights and the collision Jacobian:

$$d\tilde{\Phi} = \tilde{M}(v)\tilde{E}(I)\tilde{M}(w)\tilde{E}(J)\varphi_v(r)\psi_v(R)(1-R)R^{1/2} dR dr d\hat{\mathbf{u}}' dJ d\mathbf{w} dI dv. \quad (14)$$

If the basis is truncated at maximum degrees $K_{\max}$, $L_{\max}$, and $I_{\max}$, the total number of degrees of freedom scales as $N \propto (K_{\max} + 1)(I_{\max} + 1)(L_{\max} + 1)^2$. The resulting dense macroscopic tensor contains $\mathcal{O}(K_{\max}^3 I_{\max}^3 L_{\max}^6)$ elements. This cubic dependence on the expanded state space limits the scalability of traditional methods, motivating the exact dimensional reduction developed in Section ??.

3. Dimensional Reduction of the Polyatomic Process

This section details the reduction of the twelve-dimensional polyatomic Boltzmann collision integral to a nine-dimensional kinematic core. By integrating out the macroscopic spatial orientation over the $\text{SO}(3)$ rotation group, the intrinsic scattering and internal energy exchange processes are decoupled from the macroscopic angular geometry, establishing the Wigner–Eckart factorization for polyatomic gases.

3.1. Geometric Symmetries and Dimensionality Reduction

Evaluating the polyatomic collision operator requires integrating over a twelve-dimensional space defined by the pre-collision velocities $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$, the scattering direction $\hat{\mathbf{u}}' \in \mathbb{S}^2$, the microscopic internal energies $I, J \in [0, \infty)$, and the partition parameters $R, r \in [0, 1]$. However, the inherent isotropy of space dictates that the binary scattering kinematics possess no preferred spatial orientation. The collision process is governed by the transition probability rate B , which depends strictly on the relative speed u , the polar deflection angle χ , and the internal energy scalars.

Exploiting this rotational invariance eliminates three redundant spatial degrees of freedom. Because the internal energy variables (I, J, R, r) are purely scalar quantities, they remain invariant under any rigid-body rotation. We establish a privileged coordinate system by rigidly rotating the absolute laboratory frame to align with the colliding pair:

1. **Alignment of the target particle:** We define the local z -axis to align with the target particle's velocity, such that $\mathbf{v} = v\hat{\mathbf{z}}$. By locking $\mathbf{v}$ to the z -axis, we factor out the polar and azimuthal orientation of the collision event within the macroscopic reference frame, eliminating two spatial degrees of freedom.
2. **Alignment of the incident particle:** Due to cylindrical symmetry around the z -axis, we rotate the reference frame around $\mathbf{v}$ until the incident velocity $\mathbf{w}$ resides within the x - z plane. This integrates out the azimuthal degree of freedom of $\mathbf{w}$.

Having factored out the three-dimensional rigid-body rotation of the collision pair, the intrinsic geometry of the polyatomic elastic collision is completely described by nine independent scalar variables:

- v, w : The speeds of the target and incident particles.
- β : The polar incidence angle between $\mathbf{v}$ and $\mathbf{w}$ (constrained to the x - z plane).
- χ, ϵ : The polar deflection and azimuthal scattering angles of $\hat{\mathbf{u}}'$ relative to $\mathbf{u}$.
- I, J : The pre-collision internal energies of the target and incident particles.
- R, r : The kinetic and internal energy partition parameters.

The dimensional reduction of the polyatomic collision operator directly extends the Wigner–Eckart factorization established for monatomic gases (?). In the monatomic setting, integrating out the rigid-body rotation of the collision pair decouples the macroscopic angular geometry from a five-dimensional spatial kinematic core. For polyatomic gases, the exact same mathematical machinery applies, with two key distinctions. First, the invariant kinematic core is expanded to nine dimensions to account for the internal energy variables (I, J) and the energy partition parameters (R, r). Second, the spectral basis is augmented with orthogonal polynomials H_i acting on the internal energy space. Because these additional variables, the internal basis functions, and the polyatomic transition kernel B are strictly scalar quantities, they are intrinsically invariant under any spatial rotation $R \in \text{SO}(3)$. Consequently, the polyatomic complexity is entirely sequestered within the rotationally invariant scattering dynamics, while the macroscopic angular decoupling remains mathematically identical to the monatomic case.

3.2. Exact Factorization via Integration over the Rotation Group

Any collision configuration in the laboratory frame can be generated by applying a three-dimensional rigid rotation $R \in \text{SO}(3)$ to the privileged reference configuration. Expressing the spatial integration measure in spherical coordinates allows us to isolate the invariant Haar measure $dR = \sin \theta d\theta d\phi d\psi$ over $\text{SO}(3)$, yielding the geometric relation

$$d\mathbf{v} d\mathbf{w} d\hat{\mathbf{u}}' = dR d\Phi_{5D}, \quad (15)$$

where $d\Phi_{5D} = v^2 w^2 \sin \beta \sin \chi dv dw d\beta d\chi d\epsilon$ encompasses the five-dimensional spatial kinematic core. The total differential measure from the Cartesian formulation thus strictly decouples into the macroscopic angular measure and a nine-dimensional intrinsic measure $d\Phi_{9D} = d\Phi_{5D} d\Psi_{4D}$, with the internal measure defined as $d\Psi_{4D} = dR dr dJ dI$.

When the continuous collision geometry is rotated by R , the scalar quantities—the relative speed u , the transition kernel B , the radial Laguerre polynomials $R_{k,l}$, and the internal energy polynomials H_i —remain strictly invariant. Only the spherical harmonics transform linearly according to the Wigner D-matrices $D_{mm'}^l(R)$:

$$Y_{l,m}(\hat{\mathbf{v}}) = \sum_{m'=-l}^l D_{mm'}^l(R) Y_{l,m'}(\hat{\mathbf{v}}_0), \quad (16)$$

where the subscript 0 denotes the directional vector evaluated in the privileged frame.

Substituting the transformed spherical harmonics into the collision integral completely decouples the integration over the rigid-body rotation (dR) from the invariant nine-dimensional core ($d\Phi_{9D}$). Applying the fundamental representation identity for the product of three Wigner D-matrices over $\text{SO}(3)$, the integral over the macroscopic rotational geometry evaluates analytically to a product of Wigner 3- j symbols. Consequently, the multidimensional collision tensor separates into a sparse macroscopic angular tensor $\mathcal{G}$ and a dense, rotationally invariant physical tensor $\mathcal{R}^+$:

$$C_{\alpha_1, \alpha_2, \alpha_3}^+ = \mathcal{G}_{m_1 m_2 m_3}^{l_1 l_2 l_3} \mathcal{R}_{k_1 k_2 k_3, i_1 i_2 i_3}^{+l_1 l_2 l_3}. \quad (17)$$

The Gaunt tensor $\mathcal{G}$ encapsulates the macroscopic angular conservation laws and remains identical to the monatomic formulation:

$$\mathcal{G}_{m_1 m_2 m_3}^{l_1 l_2 l_3} = 8\pi^2 \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \quad (18)$$

The reduced physical tensor $\mathcal{R}^+$ absorbs all internal energy dependencies, the collision cross-section, and the intrinsic scattering dynamics. It evaluates the interactions strictly over the invariant 9D core:

$$\mathcal{R}_{k_1 k_2 k_3, i_1 i_2 i_3}^{+l_1 l_2 l_3} = \int_{9D} B(u, \cos \chi, I, J, R, r) P_{\text{gain}}^{l_1 l_2 l_3}(\hat{\mathbf{v}}'_0, \hat{\mathbf{v}}_0, \hat{\mathbf{w}}_0) \times R_{k_1, l_1}(v') H_{i_1}(I') R_{k_2, l_2}(v) H_{i_2}(I) R_{k_3, l_3}(w) H_{i_3}(J) d\tilde{\Phi}_{9D}. \quad (19)$$

Here, $P_{\text{gain}}^{l_1 l_2 l_3}(\hat{\mathbf{v}}'_0, \hat{\mathbf{v}}_0, \hat{\mathbf{w}}_0)$ represents the invariant geometric filter. As demonstrated in the monatomic formulation (?), this filter arises from evaluating the tensor product of the three local spherical harmonics in the privileged frame and summing over the invariant local projection indices m' . Because the polyatomic partition parameters and internal energies are strictly scalar quantities, they exert no influence on the spatial orientation of the collision vectors $\hat{\mathbf{v}}'$, $\hat{\mathbf{v}}$, and $\hat{\mathbf{w}}$. Consequently, the angular coupling mechanism remains completely independent of the internal energy exchange, and the geometric filter P_{gain} is mathematically identical to that of a monatomic gas. We refer the reader to (?) for the explicit derivation and expansion of P_{gain} via Clebsch-Gordan coefficients. Finally, the combined integration measure $d\tilde{\Phi}_{9D}$ incorporates the nine-dimensional Jacobian alongside the associated Maxwellian and internal energy reference weights:

$$d\tilde{\Phi}_{9D} = \tilde{M}(v) \tilde{E}(I) \tilde{M}(w) \tilde{E}(J) \varphi_v(r) \psi_v(R) (1 - R) R^{1/2} d\Phi_{9D}. \quad (20)$$

3.3. Incorporation of the Loss Term

The negative loss component $\mathcal{C}^-$ undergoes the identical rotational decoupling as the gain term. Because the test function and basis functions in the loss operator evaluate exclusively on the pre-collision states, their associated spherical harmonics align with the pole of the privileged frame. As detailed for the monatomic formulation (?), this configuration collapses the geometric loss filter $P_{\text{loss}}^{l_1 l_2 l_3}$ into a purely zonal mode that is entirely independent of the scattering angles.

Since both the gain and loss components share the exact same macroscopic angular Gaunt tensor $\mathcal{G}$ and nine-dimensional integration measure $d\tilde{\Phi}_{9D}$, they are merged at the continuous level to maximize computational efficiency. The total rotationally invariant collision dynamics are encapsulated in the net physical tensor $\mathcal{R} = \mathcal{R}^+ - \mathcal{R}^-$:

$$\mathcal{R}_{k_1 k_2 k_3, i_1 i_2 i_3}^{l_1 l_2 l_3} = \int_{9D} B(u, \cos \chi, I, J, R, r) R_{k_2, l_2}(v) H_{i_2}(I) R_{k_3, l_3}(w) H_{i_3}(J) \times \left[P_{\text{gain}}^{l_1 l_2 l_3}(\hat{\mathbf{v}}'_0, \hat{\mathbf{v}}_0, \hat{\mathbf{w}}_0) R_{k_1, l_1}(v') H_{i_1}(I') - P_{\text{loss}}^{l_1 l_2 l_3}(\hat{\mathbf{v}}_0, \hat{\mathbf{v}}_0, \hat{\mathbf{w}}_0) R_{k_1, l_1}(v) H_{i_1}(I) \right] d\tilde{\Phi}_{9D}. \quad (21)$$

This exact factorization reduces the $\mathcal{O}(K_{\text{max}}^3 I_{\text{max}}^3 L_{\text{max}}^6)$ complexity of the Cartesian collision tensor into a sparse $\mathcal{O}(L_{\text{max}}^5)$ macroscopic assembly coupled with the evaluation of the dense $\mathcal{O}(K_{\text{max}}^3 I_{\text{max}}^3 L_{\text{max}}^3)$ net physical tensor $\mathcal{R}$.

4. Numerical Evaluation of the Physical Tensor

In this section, we detail the numerical strategy employed to evaluate the continuous nine-dimensional integral defining the net physical tensor $\mathcal{R}$. By applying high-order tensorized quadrature rules over the invariant kinematic core, the continuous physical transition rates and internal energy exchanges are transformed into a discrete, precomputable multidimensional array. This discretization effectively bridges the continuous theoretical formulation with the highly efficient online assembly of the macroscopic collision tensor.

4.1. Quadrature Formulation and Coordinate Transformations

Evaluating the nine-dimensional net physical tensor $\mathcal{R}$ requires a highly accurate numerical integration scheme capable of resolving the polynomial basis functions, the exponential reference distributions, and the algebraic singularities inherent to the collision geometry. We achieve spectral accuracy by tensorizing optimal Gaussian quadrature rules across the invariant kinematic core.

For the five macroscopic spatial coordinates, we directly employ the singular quadrature architecture established for the monatomic Wigner–Eckart formulation (?). To resolve the non-analytic singularity of the transition kernel B

Table 1

Summary of the mapped coordinates, weight functions, and minimum quadrature bounds required to evaluate the 9D polyatomic collision core. The spatial coordinates $(z, x, y, h, \cos \chi, \epsilon)$ rely on the singularity-resolving architecture of the monatomic baseline (?). The bounds are dictated by the radial ($K_{\max}$), angular ($L_{\max}$), and internal ($I_{\max}$) spectral limits. γ denotes the interaction potential, and ν denotes the internal degrees of freedom scaling.

Coordinate	Active Domain	Quadrature Rule	Weight Function	Minimum Points (N)
Energy (z)	Both Patches	Gen. Gauss–Laguerre	$W(z) = z^{\gamma/2} e^{-z}$	$\lceil \frac{3K_{\max} + 1.5L_{\max} + 3}{2} \rceil$
Outer Frac. (x)	Patch 1 ($x > y$)	Gauss–Legendre	$W(x) = 1$	$4K_{\max} + 3L_{\max} + 4$
Outer Angle (y)	Patch 2 ($y > x$)	Gauss–Legendre	$W(y) = 1$	$4K_{\max} + 3L_{\max} + 4$
Inner Duffy (h)	Patch 1 ($y = xh$)	Gauss–Legendre	$W(h) = 1$	$K_{\max} + \lceil 1.5L_{\max} \rceil + 1$
Inner Duffy (h)	Patch 2 ($x = yh$)	Gauss–Legendre	$W(h) = 1$	$3K_{\max} + \lceil 1.5L_{\max} \rceil + 3$
Deflection ($\cos \chi$)	Both Patches	Gauss–Legendre	$W(\cos \chi) = 1$	$K_{\max} + \lceil 0.5L_{\max} \rceil + 1$
Azimuthal (ϵ)	Both Patches	Trapezoidal	$W(\epsilon) = 1$	$2K_{\max} + L_{\max} + 1$
Target Internal (I)	$[0, \infty)$	Gen. Gauss–Laguerre	$W(I) = I^{\nu} e^{-I/\tilde{T}}$	$I_{\max} + 1$
Incident Internal (J)	$[0, \infty)$	Gen. Gauss–Laguerre	$W(J) = J^{\nu} e^{-J/\tilde{T}}$	$I_{\max} + 1$
Internal Partition (r)	$[0, 1]$	Gauss–Jacobi	$W(r) = r^{\nu} (1 - r)^{\nu}$	$I_{\max} + 1$
Kinetic Partition (R)	$[0, 1]$	Gauss–Jacobi	$W(R) = R^{1/2} (1 - R)^{2\nu+1}$	$\lceil \frac{K_{\max} + 2I_{\max} + 1}{2} \rceil$

across the $v = w$ diagonal, this baseline approach rigidly rotates the domain into a center-of-mass energy variable $z \in [0, \infty)$ and applies a 2D Duffy transformation to factor out the resulting cone singularity in the fractional relative coordinates. The continuous integration is then evaluated using a combination of Generalized Gauss–Laguerre (for the energy axis), Gauss–Legendre (for the fractional parameters and polar deflection), and uniform Trapezoidal (for the periodic azimuthal angle) rules. We refer the reader to (?) for the exact polynomial bounds and selective padding strategies required to achieve spectral convergence over this 5D spatial core.

The primary numerical extension in this work is the treatment of the four polyatomic internal coordinates. These variables require specialized quadrature families to handle their distinct boundary behaviors. The associated weight functions are extracted directly from the continuous invariant measure $d\tilde{\Phi}_{9D}$:

- **Internal Energies** ($I, J \in [0, \infty)$): The integration incorporates the reference equilibrium distributions. Applying the linear scaling $x = I/\tilde{T}$, the weight function $W(x) = x^{\nu} e^{-x}$ exactly matches the canonical form for **Generalized Gauss–Laguerre** quadrature with the parameter $\alpha = \nu$.
- **Internal Partition Parameter** ($r \in [0, 1]$): The weight is dictated entirely by the partition function $\varphi_{\nu}(r) = r^{\nu}(1 - r)^{\nu}$. Mapping the domain to $[-1, 1]$ recovers the standard Jacobi weight $w(x) = (1 - x)^{\alpha}(1 + x)^{\beta}$. We evaluate this via **Gauss–Jacobi** quadrature with parameters $\alpha = \nu$ and $\beta = \nu$.
- **Kinetic Partition Parameter** ($R \in [0, 1]$): The weight combines the partition function $\psi_{\nu}(R)$ and the collision Jacobian, yielding $W(R) = R^{1/2}(1 - R)^{2\nu+1}$. Using an affine mapping to $[-1, 1]$, this is optimally evaluated via **Gauss–Jacobi** quadrature with parameters $\alpha = 2\nu + 1$ and $\beta = 1/2$.

Notably, because the asymmetric micro-reversibility quotient analytically cancels during the derivation of the weak form, the discrete physical tensor evaluates without algebraic boundary singularities. The continuous integration measure $d\tilde{\Phi}_{9D}$ natively absorbs the density of states, meaning the physical integrand evaluates to a strictly smooth and bounded polynomial sequence. This guarantees numerical stability across the entire compact manifold without requiring ad hoc truncation at $r \in \{0, 1\}$ or $R = 1$.

To maintain the spectral convergence of the Galerkin framework, the tensorized quadrature grid must resolve the algebraic degree dictated by the state truncation limits $K_{\max}$, $L_{\max}$, and the internal energy limit $I_{\max}$. The complete quadrature strategy for the fully coupled 9D polyatomic kinematic core is summarized in Table 1.

4.2. Kinematic Mapping at the Quadrature Nodes

4.3. Offline Precomputation and Online Assembly

5. Results

5.1. Validation in the Monatomic Limit

To verify that the nine-dimensional polyatomic machinery introduces no spurious cross-coupling, the operator must exactly recover the monatomic limit. This is achieved by restricting the energy partition parameters (R, r) to Dirac delta functions that strictly preserve the pre-collision internal states, and by truncating the internal energy polynomial basis to the ground state ($I_{\max} = 0$). In this degenerate limit, the numerical operator will be benchmarked against the exact Wang Chang–Uhlenbeck eigenvalue spectrum and the infinite-order Hard Sphere viscosity limit established for monatomic gases.

5.2. Spectral Convergence of the Polyatomic Quadrature

To quantify the accuracy of the singular quadrature scheme over the extended 9D polyatomic manifold, the numerical integration error will be evaluated. The l_∞ relative error of the dense physical tensor $\mathcal{R}$ will be computed against an over-resolved reference tensor and plotted against the quadrature padding parameter. This benchmark will demonstrate that the mapped Generalized Gauss–Laguerre and Gauss–Jacobi rules successfully resolve the singularities and infinite boundaries, yielding exponential spectral convergence.

5.3. Conservation Laws and the Polyatomic Null-Space

The structural stability of the discrete operator requires the exact preservation of the macroscopic collision invariants: mass, momentum, and total energy (the combined kinetic and internal energy). The operator will be evaluated against a highly non-equilibrium distribution to confirm that the numerical drift in these physical invariants remains strictly bounded by machine precision ($\mathcal{O}(10^{-15})$).

5.4. Transient Relaxation of Internal Temperatures

To explicitly validate the inelastic collision kernel, a spatially homogeneous gas will be initialized in a state of thermal non-equilibrium, where the translational temperature diverges from the internal temperature ($T_{\text{trans}} \neq T_{\text{int}}$). The temporal evolution of the non-linear system will demonstrate the correct exchange of energy between kinetic and internal modes via the R and r integrals, tracking the asymptotic relaxation toward a unified equilibrium temperature T_{eq} .

5.5. Polyatomic Transport Coefficients

A defining macroscopic feature of a polyatomic gas is the presence of bulk viscosity, which is strictly zero in the monatomic regime. By block-diagonalizing and inverting the linearized polyatomic collision operator, the infinite-order Chapman–Enskog shear and bulk viscosity coefficients will be extracted. These limits will be compared against theoretical continuous formulations (e.g., the continuous Borgnakke–Larsen model) to validate the physical fidelity of the implemented scattering mechanics.

5.6. Computational Performance and Hardware Scaling

The primary motivation for the exact dimensional reduction is computational viability. The execution time of the Wigner–Eckart factorized map-reduce algorithm will be benchmarked against a standard dense Cartesian contraction, which incurs a prohibitive algorithmic scaling of $\mathcal{O}(K_{\max}^3 I_{\max}^3 L_{\max}^6)$. Contraction timings for both shared-memory CPUs and massively parallel GPUs will be presented, highlighting the extreme efficiency gained by decoupling the evaluation of the dense physical tensor $\mathcal{R}$ from the sparse macroscopic geometric routing $\mathcal{G}$.

6. Conclusion

A. Sum-factorization and block-decoupled evaluation of the 9D core.

B. Derivation of 9D quadrature grid exactness bounds.

C. Validation Maxwell linear molecules - shear viscosity

C.1. Analytical Derivation of the Kinetic Relaxation Rates

To validate the numerical spectrum of the discretized Wang Chang-Uhlenbeck (WCU) operator, we derive the exact continuous eigenvalues for a polyatomic Maxwell gas ($\gamma = 0$) governed by the Borgnakke-Larsen collision model. We consider a purely inelastic linear polyatomic gas ($Z = 1$, $\nu = 0$) characterized by $d_t = 3$ translational degrees of freedom and $d_i = 2$ internal degrees of freedom, yielding $D = d_t + d_i = 5$ total molecular degrees of freedom. During a binary collision, the total active degrees of freedom in the center-of-mass frame is $D_{\text{coll}} = d_t + 2d_i = 7$.

C.1.1. Shear Viscosity Relaxation Rate

The shear viscosity relaxation rate is governed by the $L = 2, n = 0$ manifold of the linearized collision operator $\mathcal{L}$. The corresponding eigenfunction is the deviatoric stress tensor mode, $\phi_{\text{shear}} \propto \{\mathbf{v} \otimes \mathbf{v}\}$. Because this mode is independent of the internal energy ($i = 0$), it is unaffected by the internal energy partitions R and r . Integrating the post-collision gain term over the isotropic scattering solid angle yields an expected gain fraction of $1/2$. Subtracting the unit loss term, the exact shear viscosity eigenvalue evaluates to:

$$\lambda_{\text{shear}} \propto \left(\frac{1}{2} - 1\right) = -\frac{1}{2}. \quad (22)$$

C.1.2. Bulk Viscosity Relaxation Rate

The bulk viscosity is dictated by the thermal relaxation between the translational and internal energy pools, located in the $L = 0, n = 1$ manifold. The eigenmode orthogonal to the total energy invariant is $\phi_{\text{bulk}} \propto I - \frac{d_i}{d_t} E_{\text{kin}}$. During an inelastic collision, the expected fraction of post-collision energy deposited into the internal modes is governed by the Beta-distribution of the kinetic energy partition parameter R . The expectation value evaluates exactly to:

$$\langle 1 - R \rangle = \frac{2d_i}{d_t + 2d_i}. \quad (23)$$

Projecting the gain operator onto ϕ_{bulk} utilizing this statistical expectation yields a continuous prefactor of $d_i/(d_t + 2d_i)$. Subtracting the loss term, the bulk viscosity eigenvalue becomes:

$$\lambda_{\text{bulk}} \propto \left(\frac{d_i}{d_t + 2d_i} - 1\right) = \frac{-d_t - d_i}{d_t + 2d_i} = -\frac{5}{7}. \quad (24)$$

C.1.3. Kinetic Eigenvalue Ratio

By taking the ratio of the bulk to shear relaxation eigenvalues, the dimensional collision frequency constants cancel out entirely. For a linear polyatomic Maxwell gas, the exact analytical kinetic ratio evaluates to:

$$\frac{\lambda_{\text{bulk}}}{\lambda_{\text{shear}}} = \frac{-5/7}{-1/2} = \frac{10}{7} \approx 1.42857. \quad (25)$$

This theoretical continuum ratio serves as a rigorous, infinite-dimensional benchmark for the sum-factorized discrete collision tensor.

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Computer Code Availability

CRediT authorship contribution statement

Torsten Keßler: Conceptualization, Methodology, Software, Validation, Writing. **B.W.T. Gieling:** Methodology, Review & Editing. **Michael R.A. Abdelmalik:** Methodology, Review & Editing. **René R. Hiemstra:** Methodology, Review & Editing.